MILITARY SPECIFICATION

PROCESS OIL
(FOR USE WITH AMMUNITION)

1. SCOPE

1.1 This specification covers one type of process oil for use with ammunition (see 6.5).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109	- Inspection Terms and Definitions
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-290	- Packaging, Packing and Marking of Petroleum and Related Products

2.2 Publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

American Society for Testing and Materials ASTM

D-974	- Neutralization Number by Color-Indicator Titration
D-2270	- Calculating Viscosity Index from Kinematic Viscosity
D-88	- Saybolt Viscosity; Test for
D-97	- Test for Pour Point
D-92	- Flash and Fire Points by Cleveland Open Cup
D-270	- Sampling Petroleum and Petroleum Products
D-189	- Conradson Carbon Residue of Petroleum Products
D-1500	- ASTM Color of Petroleum Products
D-287	- API Gravity of Crude Petroleum and Petroleum Products (Hydrometer)
D-129	- Sulfur in Petroleum Products by the Bomb Method

PSC: 1375

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:46Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Materials.-The lubricating oils supplied under this specification shall be Paraffin Process Oil without additive materials.

3.2 Physical and chemical requirements.-The oil shall conform to the requirements specified in Table I.

TABLE I

Properties	Results	Test Methods
Viscosity Index, 100°F, Min.	100	4.3.1
Saybolt Universal, Seconds, 100°F, Min.	200-215	4.3.2
Pour Point °F, (Max.)	0.0	4.3.3
Flash Point, °F, Minimum, (Min.)	400	4.3.4
API Gravity	29-31	4.3.5
Carbon Residue, Percent by Weight	0.06	4.3.6
Color, ASTM, Max.	2.5	4.3.7
Sulfur Content, Percent Max.	0.05	4.3.8
Neutralization Number, Max.	0.05	4.3.9

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:46Z
Check the source to verify that this is the current version before use.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Number of units of product inspected

c. Results obtained, by defect code, for all inspections performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions.

4.2.1 Lot formation.-The term "lot" as used throughout this specification, refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. Changes to the process, specifications, or drawings, not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. The classifications of defects shall be in accordance with Standard MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:46Z
Check the source to verify that this is the current version before use.

4.2.2 Sampling.-Sampling shall be in accordance with ASTM Designation D270 If any sample fails to comply with the requirements, the lot shall be rejected.

4.3 Test methods and procedures

4.3.1 Viscosity Index, Major defect, (see 6.3), Code No. 01001.-The viscosity shall be determined in accordance with ASTM Designation D2270.

4.3.2 Viscosity Saybolt, Major defect, Code No. 02001.-The viscosity shall be determined in accordance with ASTM Designation D88.

4.3.3 Pour Point, Major defect, Code No. 03001.-The pour point shall be determined in accordance with ASTM Designation D97.

4.3.4 Flash Point, Major defect, Code No. 04001.-The flash and fire point shall be determined in accordance with ASTM Designation D92.

4.3.5 API Gravity, Major defect, Code No. 05001.-The API gravity shall be determined in accordance with ASTM Designation D287.

4.3.6 Carbon Residue, Major defect, Code No. 06001.-The carbon residue shall be determined in accordance with ASTM Desgination D189.

4.3.7 Color, ASTM, Major defect, Code No. 07001.-The ASTM color shall be determined in accordance with ASTM Designation D1500.

4.3.8 Sulfur Content, Major defect, Code No. 08001.-The sulfur content shall be determined in accordance with ASTM Designation D129.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.-Unless otherwise specified in the contract or order, packaging, packing and marking shall be in accordance with MIL-STD-290.

6. NOTES

6.1 Intended use.-The oil covered in this specification is intended for use in the manufacture of Composition C4.

6.2 Ordering date.-Procurement documents should specify the title, number and date of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:46Z
Check the source to verify that this is the current version before use.

6.3 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.4 Test results.-A copy of the test results should be forwarded to Commanding Officer, ATTN: SMUPA-ND2, Picatinny Arsenal, Dover, New Jersey, 07801.

6.5 The requirements specified in this specification are similar to the requirements of Grade 10 of Specification MIL-L-2104 except that the oil in this Specification (MIL-L-14796) does not contain any additive.

Custodian:
ARMY-MU

Preparing Activity:
ARMY-MU

Project Number: 1375-A1085

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:46Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 OCT 64 1426

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:46Z
Check the source to verify that this is the current version before use.